

Banco de validación — Detector de HR de ECG

Tiras de ECG con picos detectados vs anotaciones de cardiólogos (MIT-BIH)

`clinical.ecg_analyzer.ECGAnalyzer.detect_r_peaks()` — `prominence_factor` = 0.3 (default de la clase) vs 1.0 (candidato)

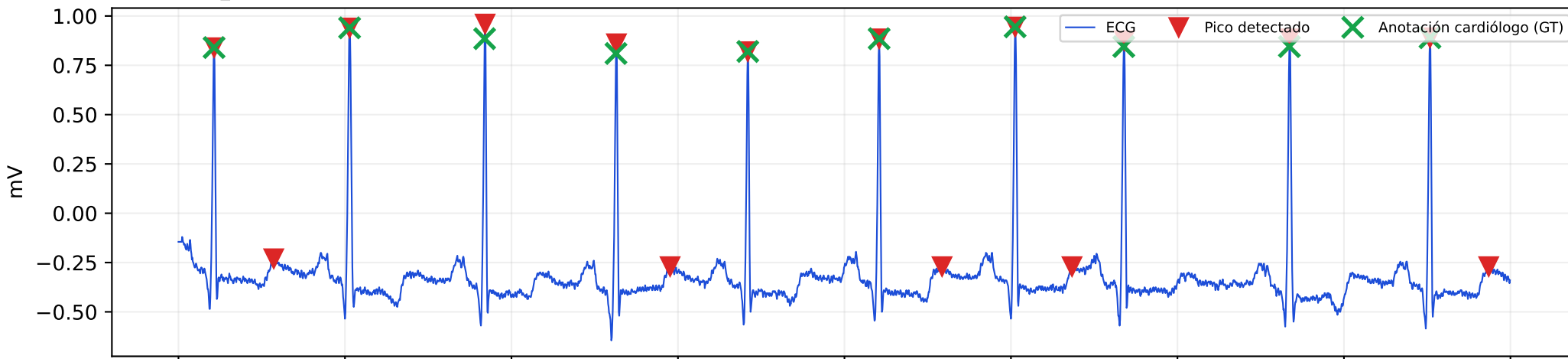
Leyenda: línea azul = señal ECG (derivación MII) · triángulo rojo = pico detectado por el algoritmo

· X verde = anotación de cardiólogo (ground truth, MIT-BIH .atr)

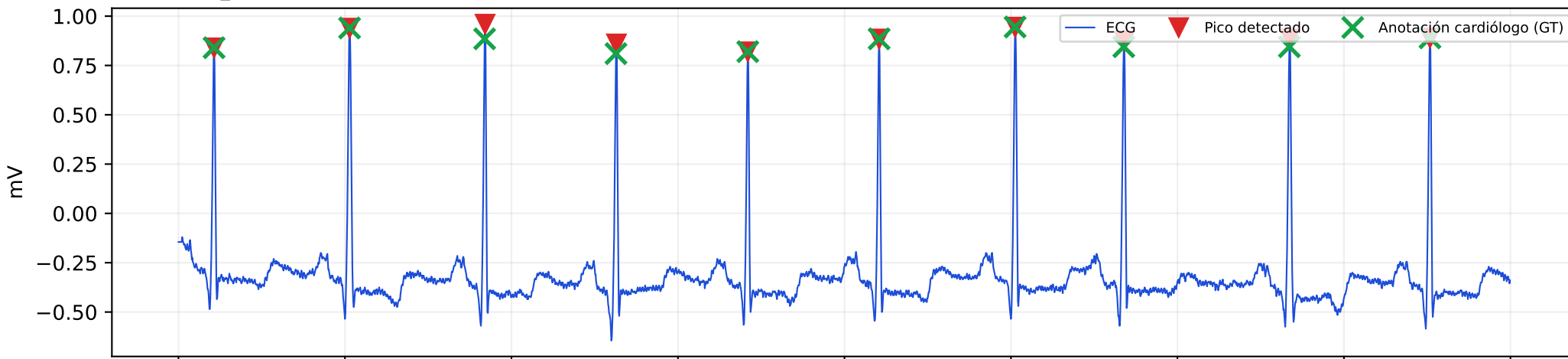
Coincidencia dentro de $\pm 50\text{ms}$ se considera acierto; un triángulo sin X cerca es falso positivo (típicamente onda T duplicada); una X sin triángulo cerca es un latido real perdido.

Registro MIT-BIH 100 — Base limpia #1 (ritmo sinusal normal, referencia clasica)
Derivación: MLII · fs=360Hz

prominence_factor = 0.3



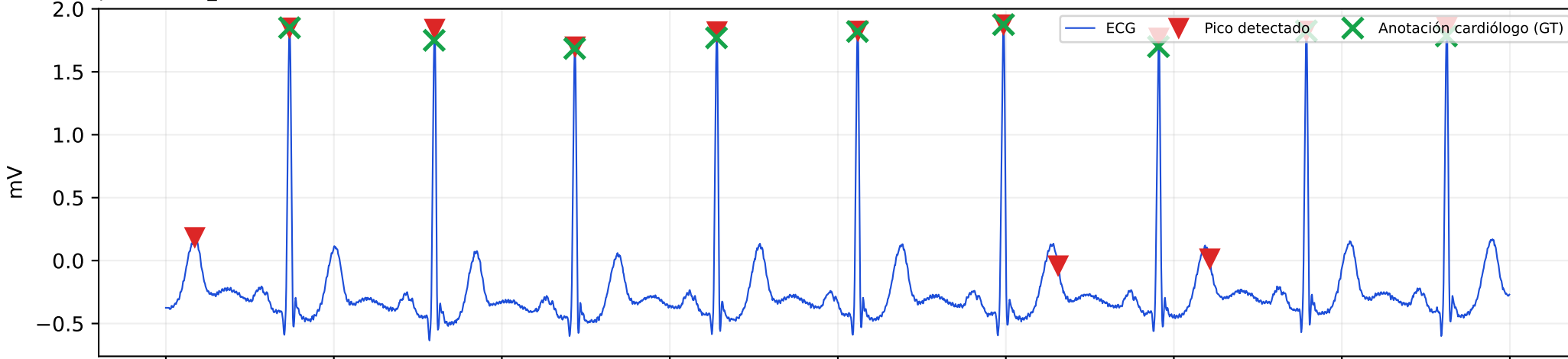
prominence_factor = 1.0



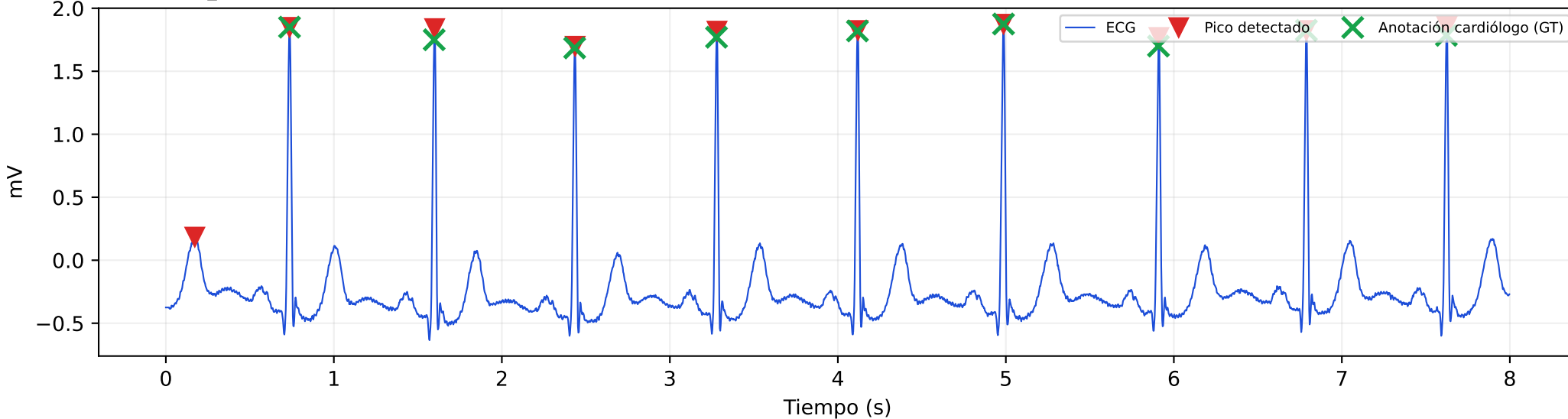
Tiempo (s)

Registro MIT-BIH 103 — Base limpia #2 (ritmo sinusal normal, morfología QRS distinta)
Derivación: MLII · fs=360Hz

prominence_factor = 0.3

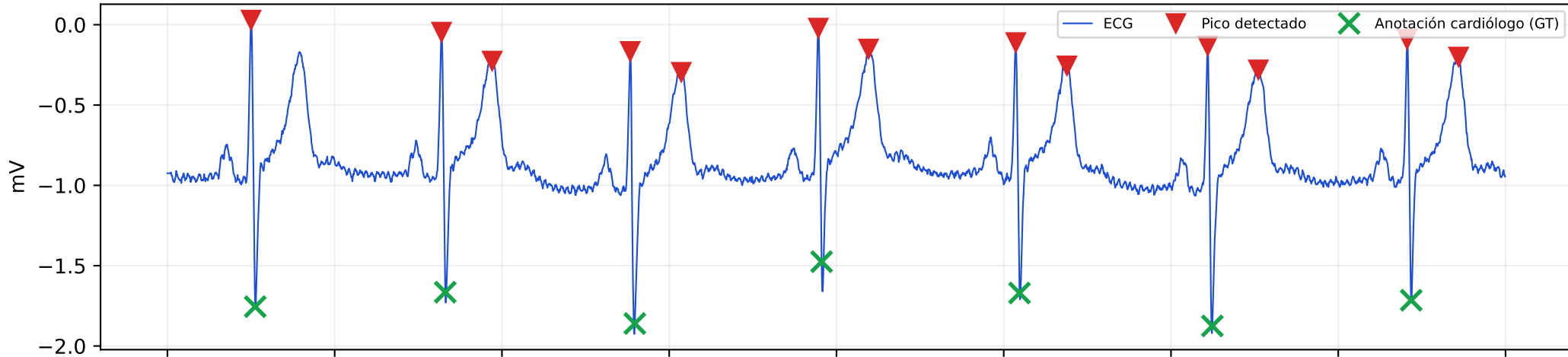


prominence_factor = 1.0

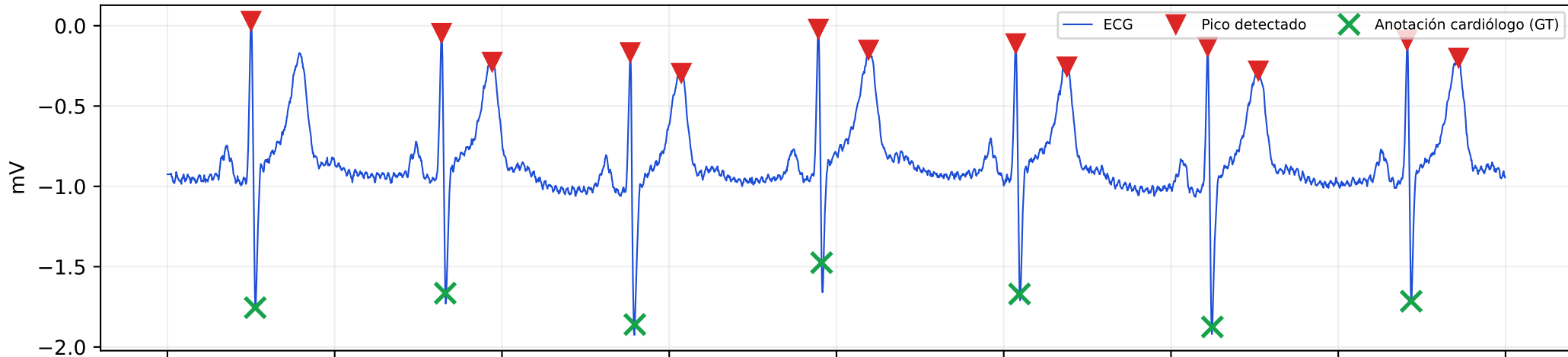


Registro MIT-BIH 117 — Trampa onda T prominente (T/R=0.79, ritmo por lo demas limpio)
Derivación: MLII · fs=360Hz

prominence_factor = 0.3



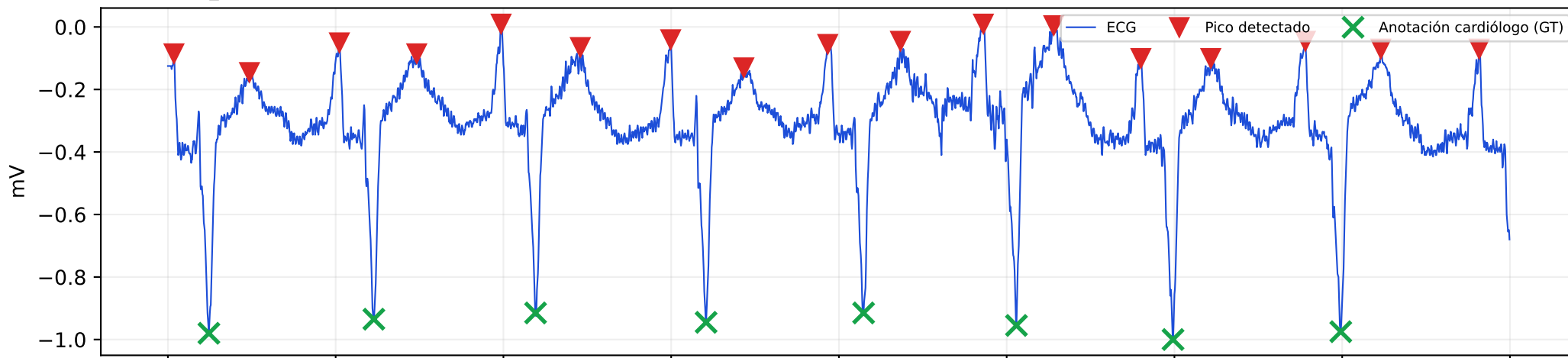
prominence_factor = 1.0



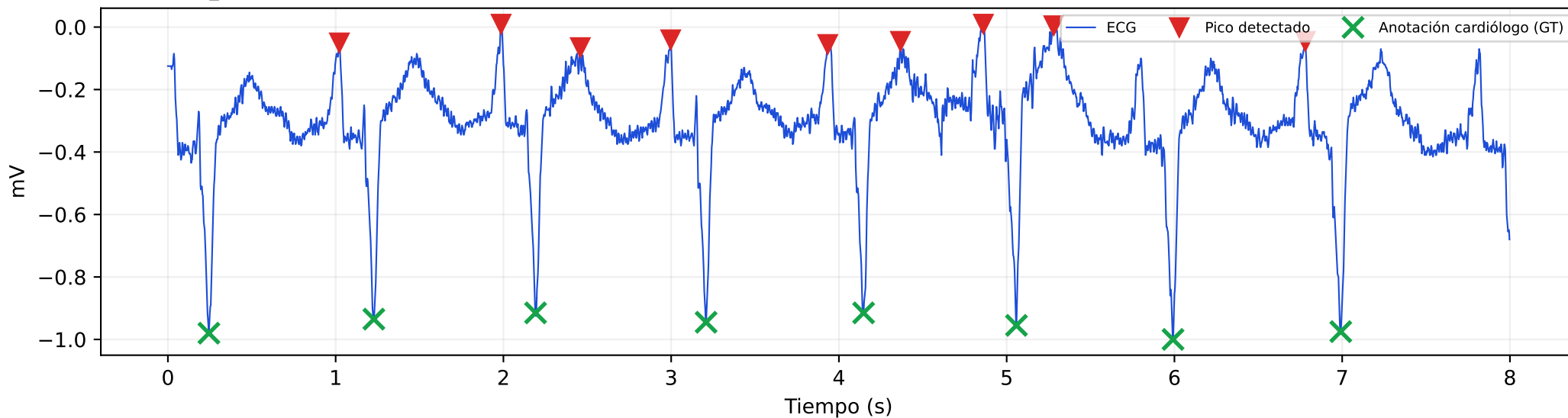
Tiempo (s)

Registro MIT-BIH 108 — Bajo voltaje QRS + PACs (qrs pico-a-pico mínimo del cribado)
Derivación: MLII · fs=360Hz

prominence_factor = 0.3



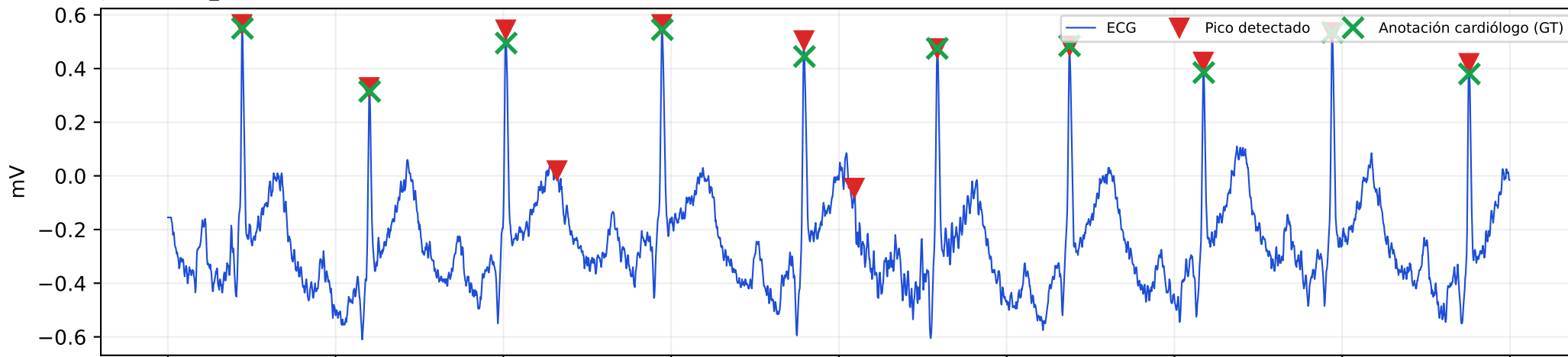
prominence_factor = 1.0



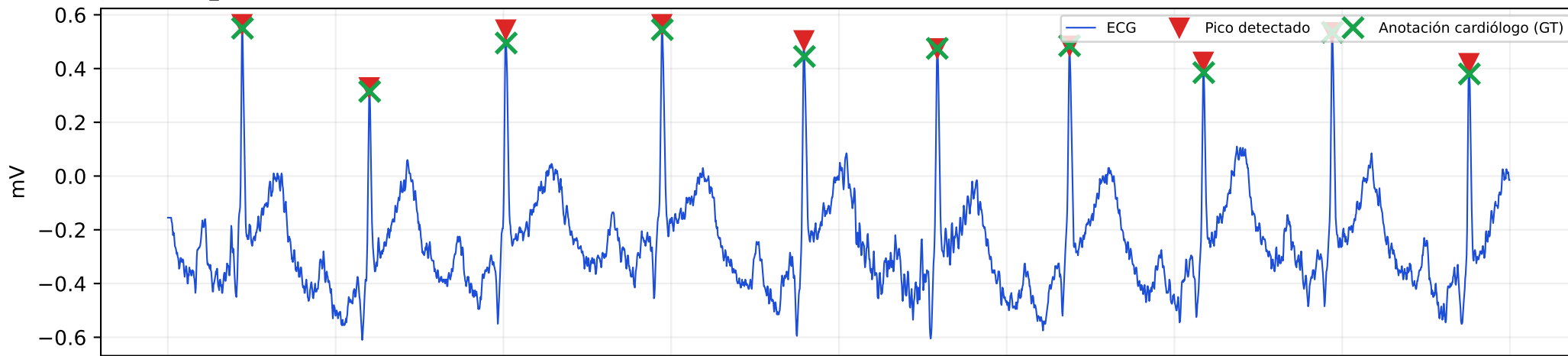
Registro MIT-BIH 228 — Arritmia real (PVCs frecuentes, rr_cv=0.385)

Derivación: MLII · fs=360Hz

prominence_factor = 0.3



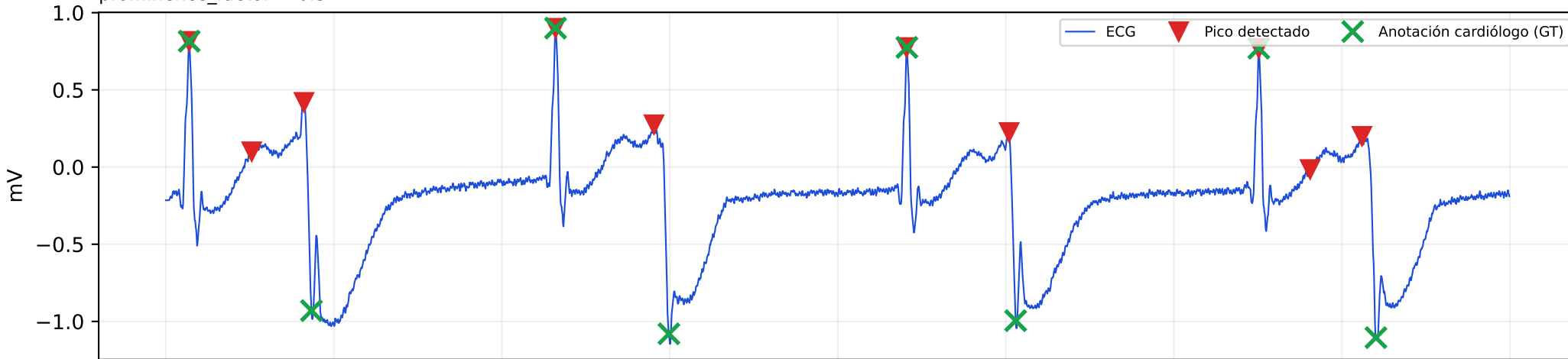
prominence_factor = 1.0



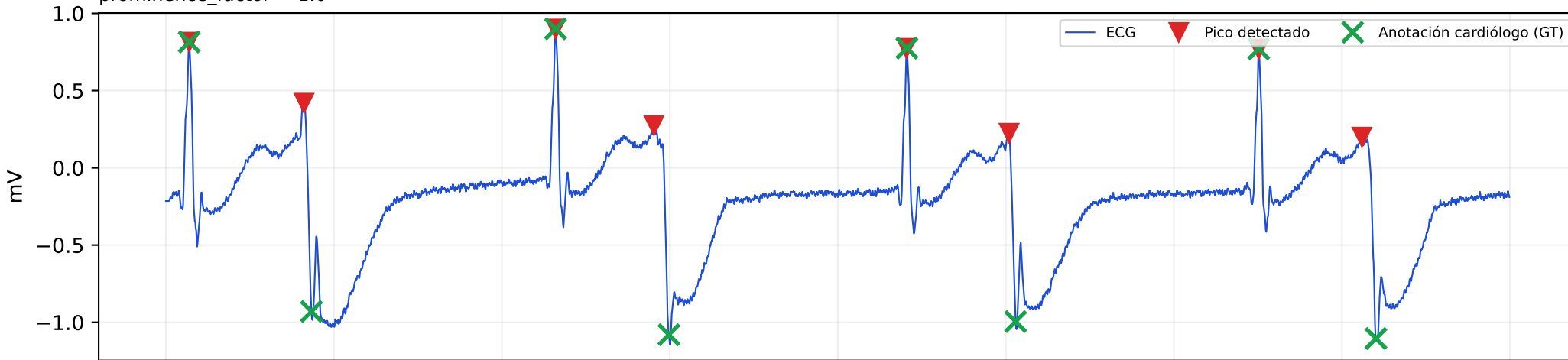
Tiempo (s)

Registro MIT-BIH 207 — Caso extremo: fibrilacion/flutter ventricular + ruido (rr_cv=0.639, el mas severo)
Derivación: MLII · fs=360Hz

prominence_factor = 0.3



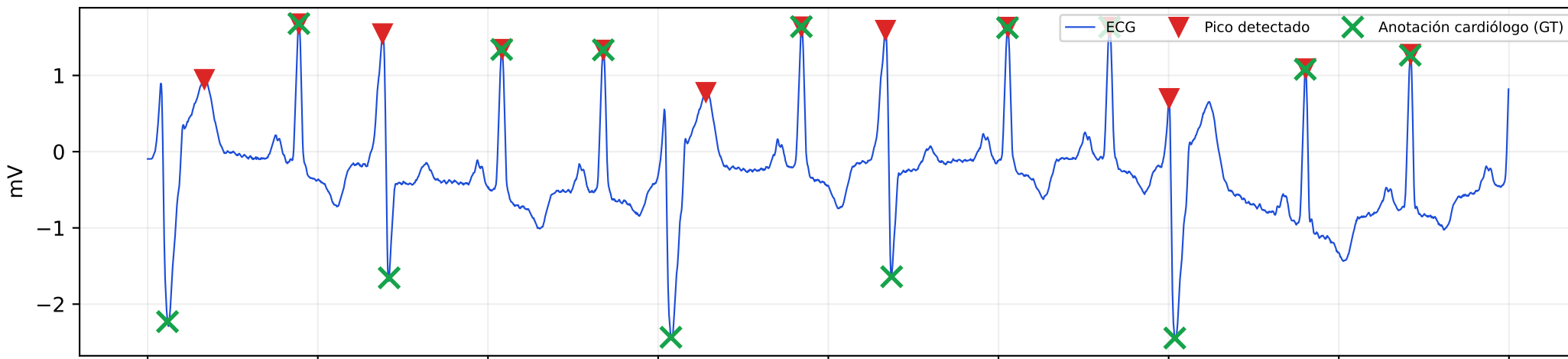
prominence_factor = 1.0



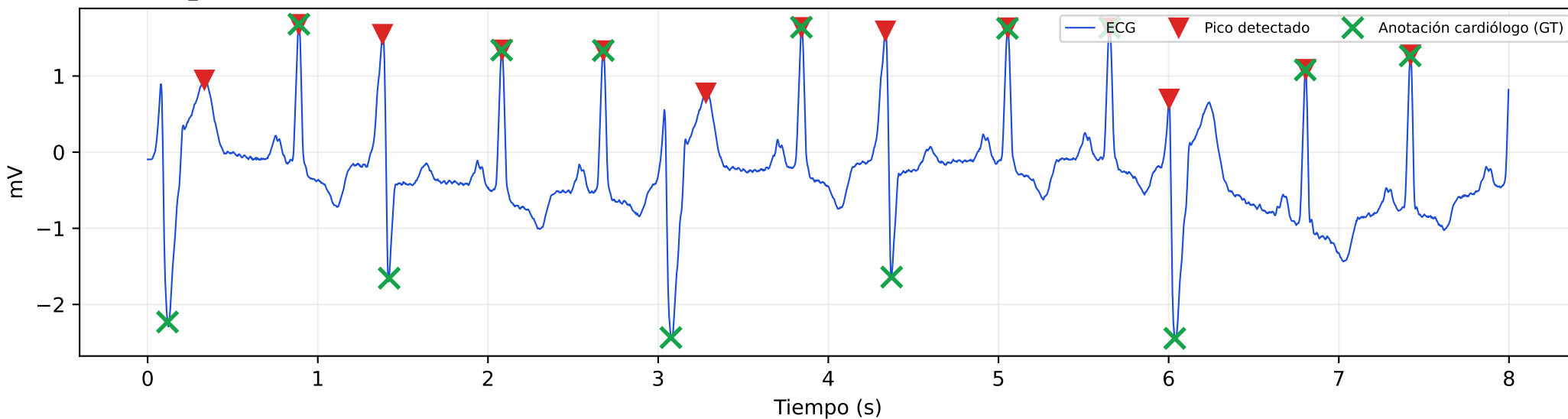
Tiempo (s)

Registro MIT-BIH 233 — Ruido/artefacto (noise_std mas alto del cribado, PVCs)
Derivación: MLII · fs=360Hz

prominence_factor = 0.3

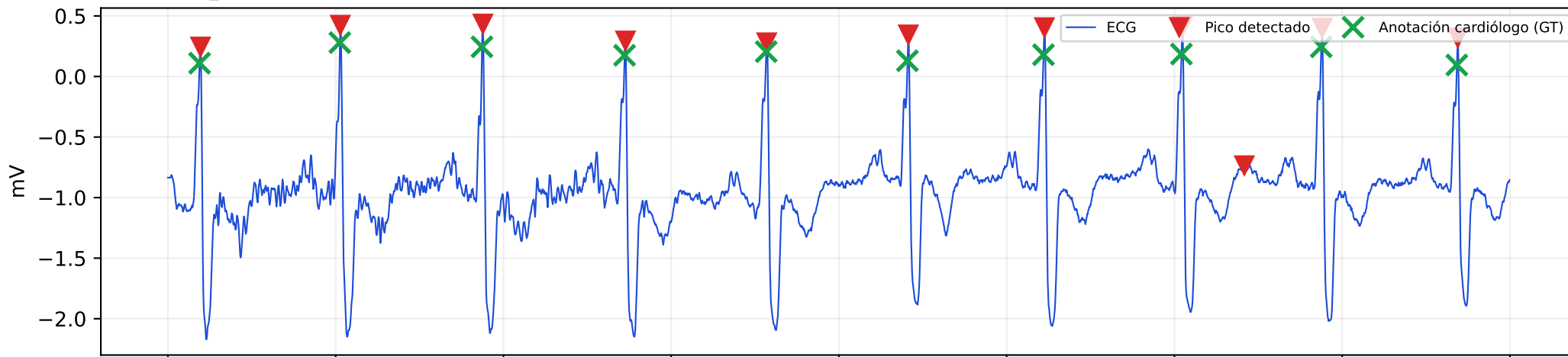


prominence_factor = 1.0

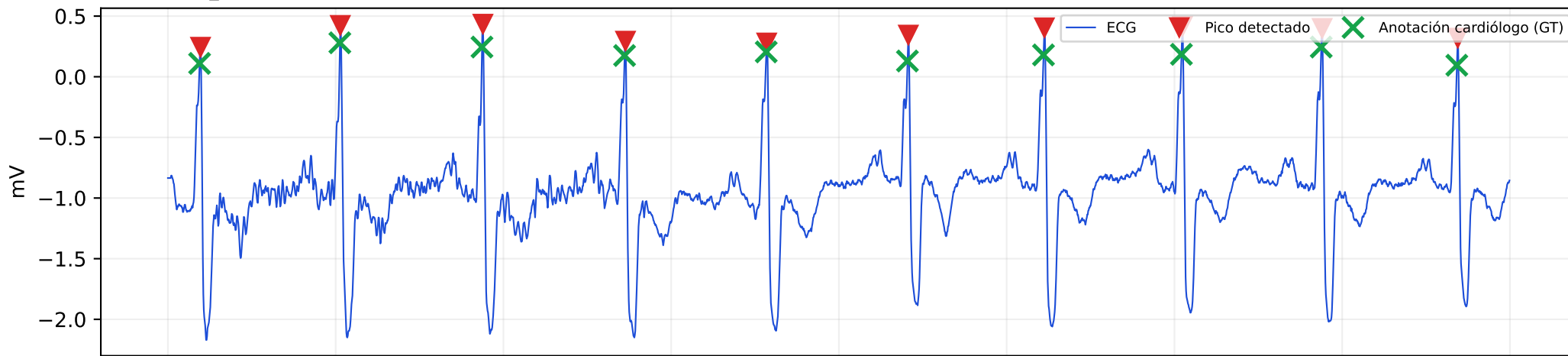


Registro MIT-BIH 118 — Bloqueo de rama derecha -- BRD (morfología QRS alterada, no arritmia de ritmo)
Derivación: MLII · fs=360Hz

prominence_factor = 0.3



prominence_factor = 1.0



Tiempo (s)